

Introduction

- Rowing performance depends on precise interaction between the blade and the water. Minor changes in blade design can affect propulsion, stroke efficiency, and blade slip. The power put into a blade differs between athletes, leaving questions about the actual changes paddle geometry effects.
- The RANDALLfoil is a newly introduced blade attachment designed to improve propulsion by reducing slip, improving force – water power transfer, and reduce the pitching of the blade during the drive phase.

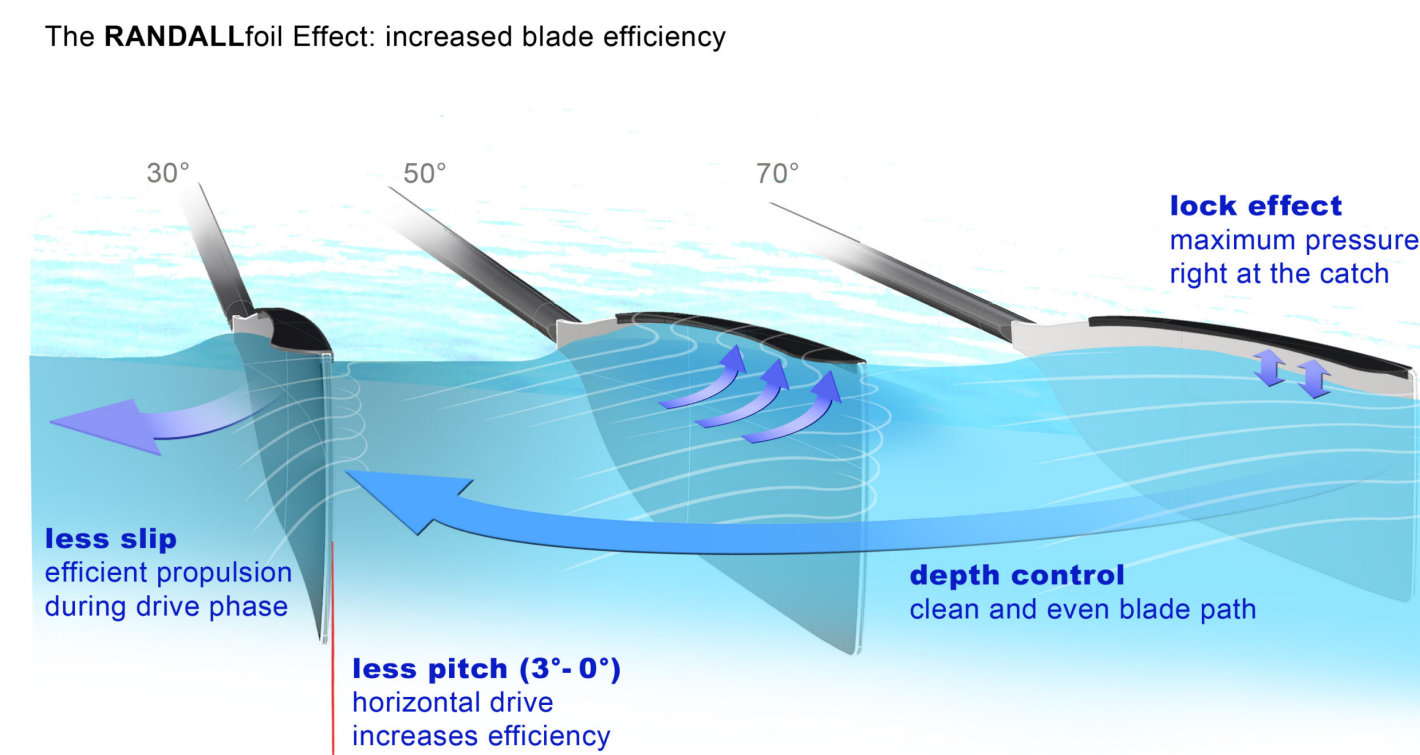


Figure 1. RANDALLfoil blade claimed benefits showing the geometry of the blade attachment and metrics along the rowing drive stroke. [1]

Objectives

- Goals of this study were to quantify the flow structures generated around rowing blades with and without the RANDALLfoil using Particle Image Velocimetry (PIV).
- Flow structures and force data was collected at 1:4 geometric and kinematic scale to compare performance between blades and vortex behavior.



Figure 2. On-water athlete (Caleb Young) shown initiating the drive phase of the motion (before blades enter the water).

Methods

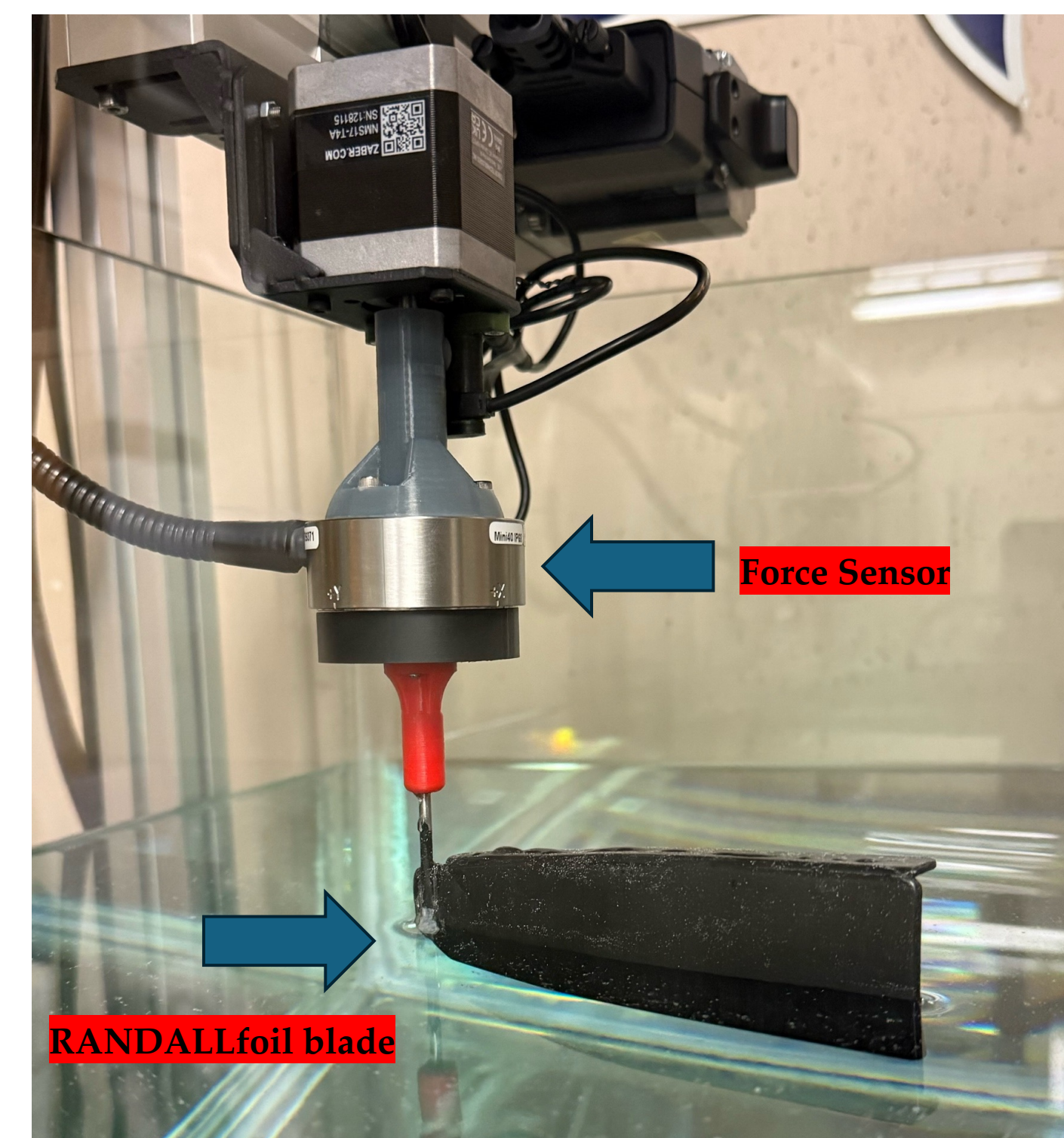


Figure 3. ATI Force-Torque sensor shown attached to a RANDALLfoil, this was the setup for force/torque data collection.

- 3D-printed rowing blade models were tested in a 24" × 24" × 24" quiescent water tank.
- Blades were mounted to linear and rotational stages that reproduced blade motion from Grift et. al. [2] at 1:4 scale of true rowing blade speed.
- ATI Nano-17/IP68 6-axis load cell Force-torque transducer was used to gather force data for both the baseline blade and the RANDALLfoil.

- Particle Image Velocimetry (PIV) was used by illuminating particles with a 532nm laser sheet. A PIV camera (LaVision MX 2M-160) was used to gather data for flow structure analysis.

PIV Data Collection

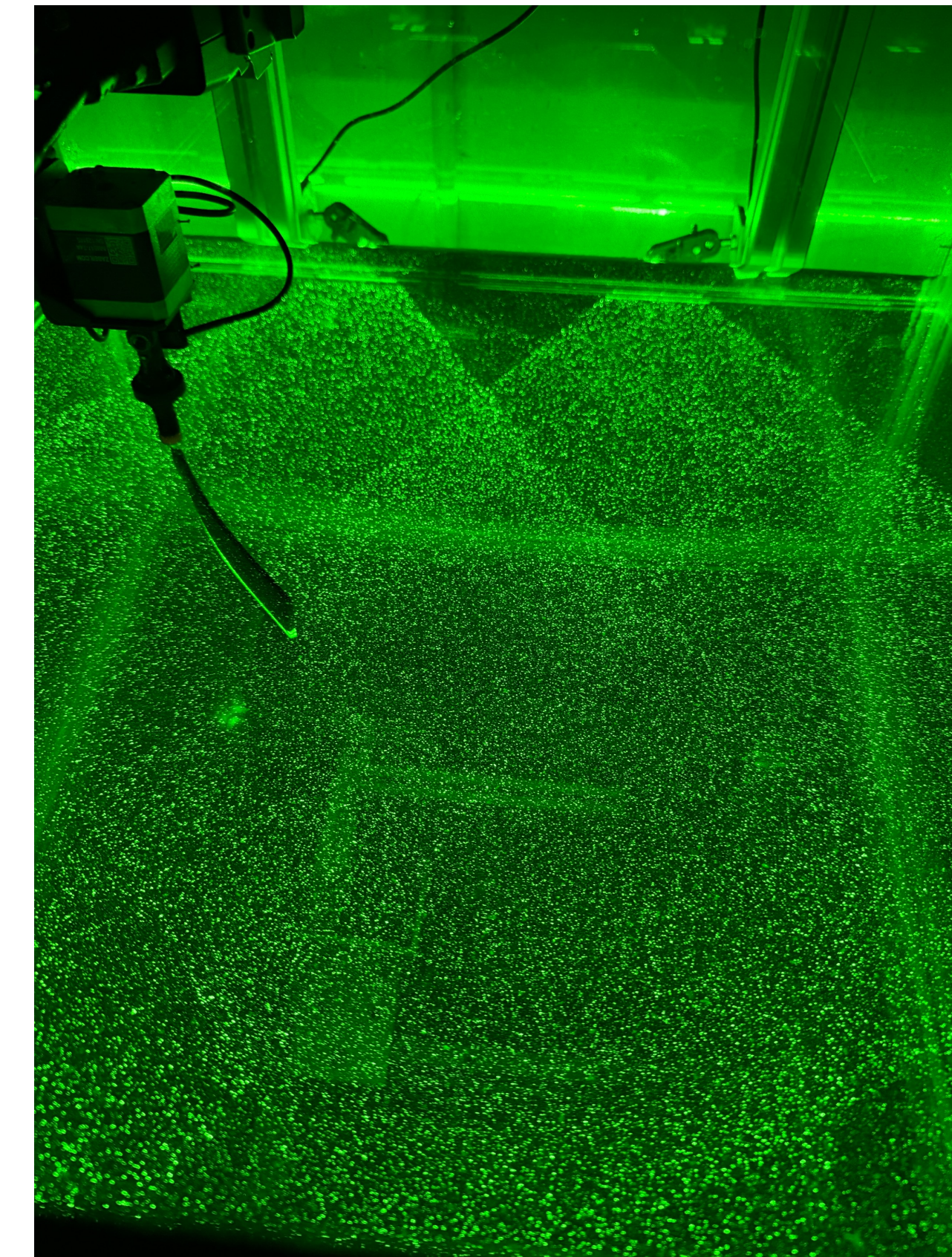


Figure 4. This shows the particles present in a quiescent tank, these are illuminated in a laser plane to get accurate PIV data.

- PIV was used to capture the flow field around each blade. Neutrally buoyant tracer particles were illuminated with a thin laser sheet, and image sequences were recorded from beneath the tank.
- Multiple fields of view were stitched together and analyzed in MATLAB to calculate velocity, vorticity, and circulation. These flow measurements were then compared with the force data to evaluate how the RANDALLfoil changed hydrodynamic performance.
- Grift et al. gave a prescribed motion path, which was set from within an excel sheet, and imported from python within Zaber controls

Flow Structure Results

- The RANDALLfoil produced stronger vortex structures than the baseline blade, particularly in the leading-edge vortex region.
- Velocity values from PIV are used to find vorticity, which is the measure of the rotation (spin) of a fluid about its center.
- PIV computes this velocity by tracking particle motion within interrogation windows across the flow field. These velocity measurements are used to calculate vorticity via spatial gradients between neighboring windows to reveal flow structures.

- A leading-edge vortex (LEV) is formed during the drive phase.
- This vortex enhances force production on the blade, as observed in the PIV results through the development of a strong coherent LEV structure, which later sheds and breaks down.

$$\omega \equiv \nabla \times v$$

- Vorticity, ω , is measured by the local velocity field from interrogation windows, which represent rotational motion.
- The pitching torque around the blade, $T_{pitch} = \int T_x dt$ is obtained from the measured force and moment data, given from the force sensor.

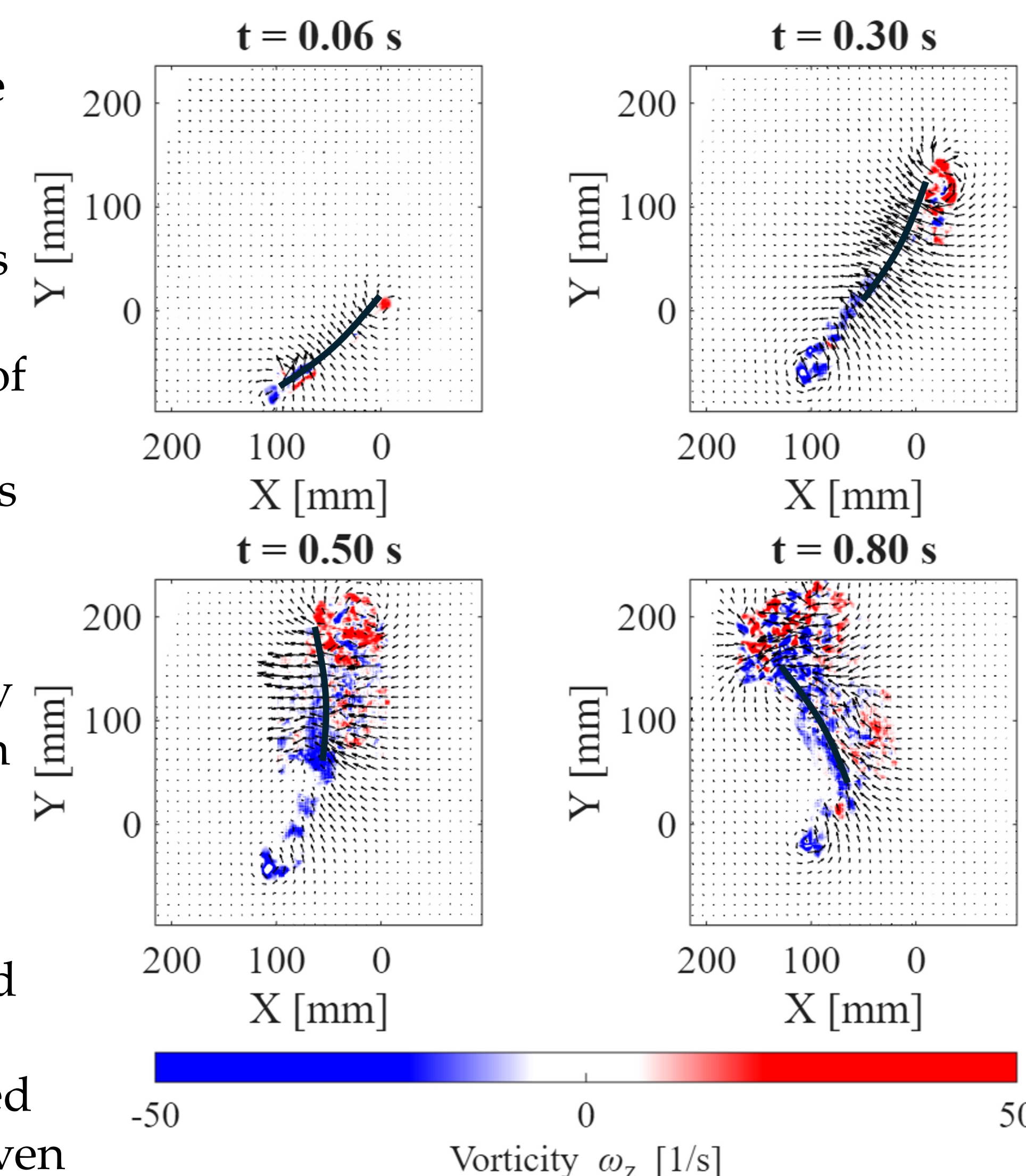


Figure 5. Vorticity plots show the blade throughout the drive motion.

Propulsion and Efficiency

- Force data was filtered using a 4th Order Butterworth filter with a 5 Hz cutoff.
- Similar filtering methods were used by Cerek et al.[5]
- These were scaled using F/κ^2 , where κ is 0.5. Using the equations below,

$$F_x = F_n \cos \theta - F_t \sin \theta \quad F_y = F_n \sin \theta - F_t \cos \theta$$

- F_x can be found, which is the main driver of forward motion.
- This is crucial for demonstrating if the RANDALLfoil shows improvement in propulsion throughout the rowing motion.

The efficiency both blades depends on propulsion. This propulsion, in the x direction, is the boat's forward motion and is defined as:

$$J_x = \int_{t_{stop}}^{t_{start}} F_x dt$$

Where is the total force factor, and E is the energy spent during the run motion that the rower would experience. Grift [2] show that this relationship can be found by finding the power, P:

$$P(t) = F(t)P(t) = F(t) \cdot V(t) + M(t) \cdot \theta(t)$$

Within this equation, M is the moment, V is the velocity vector, and $\theta(t)$ is the angular acceleration of the rowing blade.

$$E = \int_{t_{stop}}^{t_{start}} P(t) dt \quad \eta_E = \frac{J_x}{E}$$

$$\text{Percent Improvement} = \frac{\text{Randall Foil} - \text{Standard Foil}}{\text{Randall Foil}} \cdot 100$$

	Standard Foil (Mean ± STD)	Randall Foil (Mean ± STD)	Improvement (%)
J_x	0.606 ± 0.001%	0.661 ± 0.004%	8.312
E	0.074 ± 0.001%	0.053 ± 0.003%	-38.092
η_E	8.223 ± 0.120%	12.386 ± 0.670%	33.608
T_{pitch}	0.051 ± 0.0001%	0.052 ± 0.000%	1.027

Figure 7. Table above showing Force, energy, efficiency, and impulse values with standard deviation shown.

Conclusions

This study showed that the RANDALLfoil changes the flow around a rowing blade. Compared to the baseline blade, the RANDALLfoil generated stronger leading edge vortex structures and higher higher force values throughout blade motion. These results support the conclusion that the attachment improves stroke efficiency while limiting blade slip.

- [1] Randall, I. RANDALL foils. Retrieved from <https://www.randallfoils.com/>;
- [2] Grift, E. J., Tummers, M. J., & Westerweel, J. Hydrodynamics of rowing propulsion. Journal of Fluid Mechanics;
- [3] Cardoso, R., et al. Effect of Randall foils on the rowing propulsive cycle. 2024;
- [4] Li, J., & Wu, Z.-N. Unsteady lift for the Wagner problem in the presence of additional leading/trailing edge vortices. Journal of Fluid Mechanics.
- [5] Cerek, W., & Morris, S. Exploring vortex dynamics in rowing oars: Impact of the RANDALLfoil. 4th Colloquium on Vortex Dominated Flows – DisCoVor 2025, Montana State University.